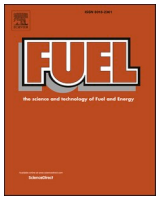
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Full Length Article



Efficient fractionation of microalgae biomass as a whole for biodiesel production enabled by deep eutectic solvent and water

Dayong Yu [a,b], Hangyu Luo [a,b], Jinyu Tan [b], Junfa Yuan [b], Jiasheng Chen [b], Xiaofang Liu [a,*], Hu Li [b,*]

[a] Guizhou Provincial Key Laboratory for Rare Animal and Economic Insects of the Mountainous Region, College of Biology and Environmental Engineering, Guiyang University, Guiyang 550005, China
[b] National Key Laboratory of Green Pesticide, Key Laboratory of Green Pesticide and Agricultural Bioengineering, Ministry of Education, State-Local Joint Laboratory for Comprehensive Utilization of Biomass, Center for R&D of Fine Chemicals, Guizhou University, Guiyang, 550025, China

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ABSTRACT

The fractionation of microalgae biomass and its efficient utilization have always been problems to be solved. Herein, microalgae biomass was pretreated with deep eutectic solvent (DES) to fractionate lipids, proteins, and carbohydrates, which were further converted into biodiesel. DES (CC-LA) synthesized from lactic acid and choline chloride had high fractionation efficiency, and the lipid extraction rate (17.7 %) was 2.3 times higher than that of the untreated one (7.7 %). Adding 0.5 mol water in CC-LA changed the polarity of the solvent, and the hydrogen bond acidity (α) and hydrogen bond alkalinity (β) increased from 1.37 and 0.52 to 1.42 and 0.54, respectively. Thus, the separation efficiency of DES was significantly improved. The lipid extraction rate was as high as 89.4 %, the protein dissolution rate was as high as 84.2 %, and the carbohydrate saccharification rate was as high as 96.8 %. In addition, the isolated protein and glycosylated carbohydrates could be converted into 8.9 g microbial lipids/100 g microalgae biomass by *Lipomyces starkeyi*. The new water-promoted DES fractionation technology has dramatically enabled microalgae biomass's efficient and comprehensive utilization.

1. Introduction

Microalgae are unicellular organisms that are easy to culture, can efficiently fix CO_2 and reproduce rapidly, and usually contain more than 80 % organic matter, such as lipids, proteins, and carbohydrates [1]. These substances are high-quality sources for the preparation of biofuels. Therefore, microalgae are considered the new generation (third generation) of biomass raw materials for biofuel production [2,3]. Biofuels based on microalgae biomass mainly include bioethanol (fermentation of carbohydrates), biodiesel (transesterification of lipids), and biomethane (anaerobic fermentation) [4]. However, the utilization of microalgae biomass faces the problem of low efficiency. Because a large number of available substances are stored in the cytoplasm, the high resistance cell wall of microalgae hinders their effective utilization [5]. In addition, in most cases, only carbohydrates and lipids are used, and a large amount of protein is discharged as waste, resulting in a loss of overall biomass utilization [6]. To address this, it is crucial to develop an efficient and green pretreatment technique to efficiently pretreat and fractionate microalgae biomass (e.g., lipids, carbohydrates, and proteins), and the total biomass was converted into biodiesel to efficiently utilize microalgae biomass.

The main technical problems for microalgae valorization are the destruction of the microalgae cell wall and the fractionation and utilization of total biomass. Currently, pretreatment methods include mechanical pretreatment (e.g., milling, high-pressure homogenization, microwave, ultrasonication, and pulsed electric field), chemical pretreatment (e.g., acid, alkali, organic soils, and oxidant), and biological pretreatment (e.g., enzymes and microorganisms) [7,8]. However, several problems are constantly encountered, such as high-energy consumption, corrosion of equipment, destruction of the environment, and inability to fractionate microalgae biomass effectively. It is necessary to explore a green and economical pretreatment/fractionation technology. Therefore, many new green solvents have been developed for pretreatment, such as ionic liquid (IL), liquid biphasic system (LBS), and deep eutectic solvent (DES).

DES and IL have similar properties, such as non-flammable, non-volatile, recyclable, low vapor pressure, melting point, thermal stability, and so on [9]. However, compared with IL, DES has the advantages of

* Corresponding authors.
E-mail addresses: liuxfzap@163.com (X. Liu), hli13@gzu.edu.cn (H. Li).

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low cost, simple preparation, non-toxicity, and biodegradability, which makes it more promising [10]. LBS is used for the pretreatment of microalgae biomass, which has cost saving advantages, high separation efficiency, and protein invariance. However, the premise of its practical implementation is the need to destroy the cell wall of microalgae in advance [11]. Studies have shown that DES can effectively destroy the hydrogen bond network on the cell wall of microalgae to achieve the purpose of cracking microalgae cells.

Meanwhile, DES has a high affinity for proteins and can quickly dissolve and extract a large number of proteins during the treatment of microalgae [12]. Significantly, the composition of DES can be freely adjusted to change its physicochemical properties. DES is hydrophilic, and adding an appropriate amount of water can reduce its viscosity and significantly improve its pretreatment performance [13,14]. It is, hence, an excellent choice to prepare suitable DES to pretreat fractionated microalgae biomass.

Upon pretreatment of microalgae biomass with DES, lipids are converted into biodiesel by transesterification after extraction, and carbohydrates can be converted into biofuels by fermentation. Therefore, the fate of proteins dissolved in the DES phase must be considered. *Lipomyces starkeyi* (*L. starkeyi*) is an excellent oleaginous yeast which produces microbial lipids that can be converted into biodiesel by transesterification [15].Notably, the substrates required for fermentation are broad-spectrum, including reducing sugars, starch, lignocellulosic hydrolysates, proteins, and fatty acids [16]. Therefore, the fractionated carbohydrates and proteins in the DES phase can be used as substrates for *L. starkeyi* to produce microbial lipids, thereby efficiently utilizing whole microalgae biomass. At the same time, it is necessary to consider the toxic effect of DES components on microorganisms during fermentation. Choline chloride-based DES has good biocompatibility and is widely used in biotransformation [17,18]. Therefore, choline chloride (CC) was used as a hydrogen bond acceptor to prepare different DESs in this paper.

To fully utilize microalgae biomass, the concept of “full utilization” for producing biodiesel would be a promising strategy. In this work, a sustainable DES composed of CC and lactic acid (LA) with efficient pretreatment performance was screened out, and the effect of water on the pretreatment performance of the DES was explored. The influences of temperature and time on the optimal DES fractionation efficiency were investigated. After the reaction, the microalgae lipids were obtained by fractionation, and the proteins were fermented as substrates to convert into microbial lipids. The remaining solids (carbohydrates) were enzymatically hydrolyzed into monosaccharides as fermentation substrates, which were also converted into microbial lipids. Finally, the harvested microalgae lipids and microbial lipids were converted into biodiesel. The fractionation and “full utilization” of microalgae biomass components provide a good reference for upgrading biomass to biofuels or value-added chemicals.

2. Materials and methods

2.1. Microalgae and chemicals

Microalgae powder (*Nannochloropsis* sp.) was purchased from Shandong Wudi Lvqi Biological Engineering Co., ltd. *Nannochloropsis* sp. was dried at 105 °C for 48 h to calculate its water content. Carbohydrates, protein, and lipids of microalgae were determined by the National Renewable Energy Laboratory (NREL/TP-5100–60957) standard technique [19], Lowry method [20], and Bligh & Dyer method [21], respectively. The content of each component are water 10.5 ± 0.2 %, carbohydrate 15.3 ± 0.4 %, protein 39.2 ± 0.3 %, lipid 30.3 ± 0.6 %, and ash 11.1 ± 0.4 % respectively. The dried microalgae powder was stored in a sealed container and placed in a desiccator for later use. CC (≥99.0 %), formic acid (FA, ≥98.0 %), acetic acid (AA, ≥99.5 %), LA (≥99.0 %), and other chemicals of analytical purity grade were acquired from InnoChem Science & Technology Co., Ltd. (Beijing, China).

2.2. Dess synthesis and pretreatment

Hydrogen bond acceptor (HBA) (i.e., CC) and hydrogen bond donor (HBD) (FA, AA, or LA) in a molar ratio of 1:2 were stirred at 80 °C and 600 rpm for 30 min to obtain a clear and transparent colorless liquid, denoted as CC-FA, CC-AA, and CC-LA [12]. For microalgae pretreatment in a typical procedure, CC-LA was added together with 0.5, 1.0, and 2.0 mol of water. Before use, the prepared DESs were stored in sealed containers and placed in a dryer to avoid absorbing moisture in the air. Typically, a mixture of 0.5 g microalgae powder and 10 g DES was placed in a pressure tube and heated at 600 rpm for pretreatment. The set heating temperature was 80, 100, 120, and 140 °C, and the required time was 2, 4, 6, 8, and 10 h.

2.3. Extraction of microalgae lipids and enzymatic hydrolysis of residual carbohydrates

After pretreatment of microalgae with DES, 20 mL water was added to the reaction system, and then an equal volume of ethyl acetate was added to extract lipids. The upper ethyl acetate was harvested by centrifugation, and the above steps were repeated 3 times. The collected ethyl acetate was combined to remove the solvent by rotary evaporation, and the obtained microalgae lipids were placed at 105 °C to remove water and weighed to calculate the lipid extraction rate. The lipid extraction rate was determined by following the below equation:

$$\text{Lipid extraction rate}\,(\%) = \frac{\text{Weight of lipid extracted}}{\text{Lipid weight in initial microalgae}} \times 100\% \quad (1)$$

The DES aqueous phase was recovered to detect the content of monosaccharides and proteins and used for subsequent fermentation. The remaining solids were douched 3 times with double distilled water (ddH_2O) to remove the residual DES and then dried at 105 °C to obtain the recovered microalgae powder (RMP) for subsequent enzymatic hydrolysis.

The enzymatic hydrolysis reaction was evaluated under a 50 mM phosphate buffer (pH 4.8) with a solid (raw and RMP) loading of 1 % (w/v) and 20 FPU/g solid cellulase (Cellic® CTec2, Filter paper activity: 150 FPU/mL, Novozyme). A final concentration of 50 μg/mL tetracycline was then added to inhibit bacterial growth. The system was placed in a 50 mL Erlenmeyer flask and then hydrolyzed at 200 rpm for 12 h in a preheated (50 °C) shaker. The hydrolysate was taken out at intervals, and the concentration of monosaccharide was determined after the inactivation of the enzyme at 100 °C for 10 min. The concentrations of monosaccharide was detected by HPLC (Agilent 1260, Hi-Plex H, 300 × 7.7 mm, 8 μm) equipped with a differential refractive index detector (RID) with 0.05 mol/L dilute H_2SO_4 as the mobile phase (0.5 mL/min). The temperature of the column and RID was kept at 50 ∘C. The monosaccharide yield was determined following the equation:

$$\text{Monosaccharide yield}\,(\%) = \frac{\text{Monosaccharide in the enzymatic liquid}\,(\text{g})}{\text{Monosaccharide in the substrate}\,(\text{g})} \times 100\% \quad (2)$$

2.4. Strain and cultivation

Lipomyces starkeyi 2.2291 (*L. starkeyi*), supplied by China General Microbiological Culture Collection Center (CGMCC), was cultured on YPD (yeast peptone dextrose) medium (yeast extract: 10 g/L, tryptone: 20 g/L, glucose: 20 g/L and preparation of solid medium with additional agar: 20 g/L) and stored at 4 °C. The medium was sterilized in an autoclave at 115 °C for 20 min, and the glucose solution was sterilized alone.

The general culture procedure of *L. starkeyi* includes the following steps. (1) A pure culture of *L. starkeyi* was obtained according to the

activation procedure provided by CGMCC. (2) The pure culture was inoculated on a YPD solid plate, and a single colony was obtained after 48 h at 28 °C. (3) The single colony was inoculated in a YPD liquid medium, and the enriched pure culture was obtained after 24 h at 28 °C as the initial culture for fermentation.

2.5. Fermentation of DES aqueous phase after microalgae pretreatment

To test the fermentability of the DES aqueous phase recovered after DES pretreatment of microalgae, *L. starkeyi* (microbial lipid production by fermentation) was used to evaluate the feasibility. The recovered DES aqueous phase was adjusted to pH 6.0 using solid NaOH, and then 0.5 g/L $MgSO_4{\cdot}7H_2O$ and 1.0 g/L KH_2PO_4 were added. After sterilization at 121 °C for 20 min, it was used as a medium for *L. starkeyi* fermentation. The initial culture of *L. starkeyi* after YPD culture was centrifuged to discard the medium, and the cells were washed with sterile ddH_2O 3 times and resuspended, followed by addition to the medium prepared with DES aqueous phase in an inoculation amount of 10 % (v/v) so that the cell concentration in the initial fermentation broth was $OD_{600} = 1.0$ [22]. Subsequently, the fermentation system (in a 250 mL Erlenmeyer flask) was cultured in a shaker at 28 °C and 200 rpm for 72 h, and samples were taken every 12 h to determine the consumption of monosaccharides and proteins and the yield of dry cell weight (DCW) and microbial lipid.

2.6. Fermentation of RMP enzymatic hydrolysate

To meet the high C/N ratio required for *L. starkeyi* fermentation, the solid loading for RMP hydrolysis was adjusted to 5 %. The hydrolysate obtained by replacing RMP with untreated, dry microalgae powder was used as a check (CK). 0.5 g/L $MgSO_4{\cdot}7H_2O$, 1.0 g/L KH_2PO_4, and 0.22 g/L $(NH_4)_2SO_4$ were added to the obtained hydrolysate and then sterilized at 121 °C for 20 min and stored in a 250 mL Erlenmeyer flask. Subsequently, *L. starkeyi* was inoculated as described in Section 2.5 and fermented at 28 °C and 200 rpm for 72 h. Sampling was performed every 12 h to determine monosaccharide consumption (DCW) and microbial lipid production.

2.7. Lipid extraction

After the fermentation broth was harvested, the wet cells were obtained by centrifugation, washed with ddH_2O 3 times to remove the residual medium on the cell surface, and then freeze-dried to obtain dry cells. Lipid extraction was performed, according to the published literature [23]. The dry cells were resuspended with 4 M HCl to make the cell suspension concentration (0.1 g/mL). After standing at room temperature for 1 h, it was shaken well again and placed in a water bath at 100 °C for 10 min to lyse the cells fully. Then, an equal volume of methanol and 2 times chloroform were added, fully oscillated, and mixed for 30 min, followed by centrifuging at 8,000 rpm for 5 min. The chloroform layer was recovered, and the same volume of chloroform was added again, repeated 3 times. All chloroform was collected, and the solvent was removed by rotary evaporation. The obtained microbial lipid was dried to constant weight at 105 °C. DCW and lipid yield were determined using equations (3) and (4).

$$\text{DCW (g/L)} = \frac{\text{Weight of dried cell}}{\text{Volume of fermentation medium}} \tag{3}$$

$$\text{Lipid yield (g/L)} = \frac{\text{Weight of extracted lipid}}{\text{Volume of fermentation medium}} \tag{4}$$

2.8. Lipid transesterification and fatty acid methyl esters (FAMEs) composition analysis

According to previous studies [24], a lipid transesterification reaction was completed in the methanol-sulfuric acid system. Typically, the extracted microalgae and microbial lipids were dissolved in methanol and transferred to a 15 mL pressure tube, then concentrated sulfuric acid was added to a final concentration of 2 % (w/w). The reaction system was placed at 80 °C for 2 h, and an equal volume of ddH_2O and an equal volume of ethyl acetate were added to thoroughly mix. Subsequently, the composition of FAME was analyzed by GC–MS (Agilent 6890 N GC, 5973 MS) and GC (Agilent 7890B, HP-5 column: 30 m × 0.32 mm × 0.25 μm, and flame ionization detector) with SUPELCO standard mix as reference.

2.9. Monosaccharides analytic

After pretreatment, enzymatic hydrolysis, and fermentation broth, the monosaccharides in DES aqueous phase were analyzed by HPLC (Agilent 1260, Hi-Plex H, 300 × 7.7 mm, 8 μm). The mobile phase was 50 mM H_2SO_4 (0.5 mL/min), the detector was RID, and the column temperature and detector temperature were set at 50 °C.

2.10. Characterization of DES and RMP

2.10.1. Characterization of DES

Fourier transform infrared (FT-IR) spectra of DESs were evaluated on Nicolet IS50 AEM (Thermo Fisher) with wavenumbers ranging from 4000 to 400 cm^{-1}. The dipole/polarizability (π*), hydrogen bond acidity (α), and hydrogen bond alkalinity (β) of DESs solvents were determined by Kamlet-Taft (K-T) solvatochromic parameters [25,26]. α, β, and π * values were calculated according to the following equations:

$$\pi^* = 0.314 \times (27.52 - \hat{I}\frac{1}{2}\ \text{NEt2}) \tag{5}$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \hat{I}\frac{1}{2}\ \text{NR}}{1.6078} \tag{6}$$

$$\beta = 11.134 - \frac{3580}{\hat{I}\text{»}(\text{NH2})\text{max}} - 1.125\pi^* \tag{7}$$

$$\hat{I}\frac{1}{2}\ \text{NR} = \frac{104}{\hat{I}\text{»}(\text{NR})\text{max}} \tag{8}$$

where ν_{NEt2}, $\lambda(NH_2)_{max}$, and $\lambda(NR)_{max}$ represent the maximum absorption wavelength of N,N-dimethyl-4-nitrobenzene, 4-nitroaniline and Nile Red, respectively.

2.10.2. Characterization of RMP

FT-IR spectra of RMP were evaluated on Nicolet IS50 AEM (Thermo Fisher) with a KBr disk in a wavenumber range of 4000 to 400 cm^{-1}. The scanning electron microscope analyzed raw material and RMP surface morphology (SEM, ZEISS SIGMA300). The N_2 adsorption–desorption isotherms of RMP were performed on a Micromeritics ASAP 2460 analyzer (196 °C) to measure the Brunauer-Emmett-Teller (BET) surface area. Raw materials and RMP were recorded using X-ray diffraction (XRD) at 2θ degree of 10° to 80° using a Bruker D8 Advance X-ray powder diffractometer with Cu Ka (λ = 1.542 Å). Furthermore, the crystallinity index (CrI) of the sample was calculated using the below CrI formula.

$$\text{CrI (\%)} = \frac{(\text{I002-Iam})}{\text{I002}} \times 100\% \tag{9}$$

Where I_{002} is the maximum intensity of the crystalline region, and I_{am} is the intensity of the amorphous region (baseline) [27].

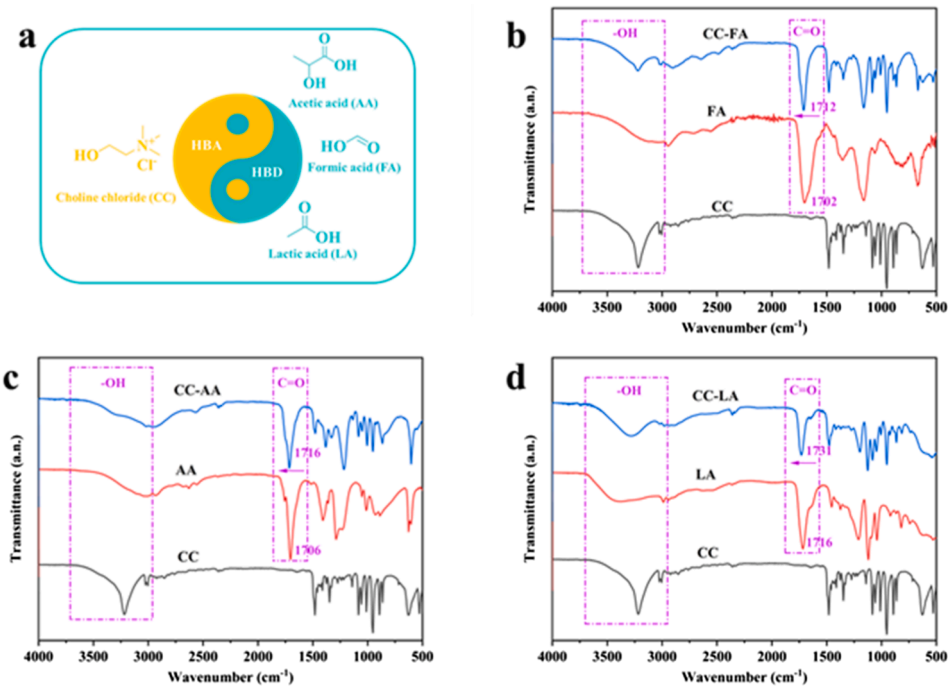


Fig. 1. (a) Chemical structures of HBDs and HBA for DESs synthesis; (b-d) FT-IR spectra of pure components and DESs.

Table 1
The Kamlet-Taft parameters of DES and the BET surface area, and CrI of RMP.

NO.	DES	Kamlet -Taft parameters			BET Surface Area (m^2/g)	CrI (%)
		π*	α	β		
Raw	–	–	–	–	0.66	72.3
1	CC-FA	1.18	1.35	0.47	1.44	80.4
2	CC-AA	0.99	1.19	0.51	0.98	77.1
3	CC-LA	1.16	1.37	0.52	13.02	83.2
4	CC-LA-$W_{0.5}$	1.19	1.42	0.54	19.62	85.4
5	CC-LA-W_1	1.15	1.36	0.51	10.86	82.3
6	CC-LA-W_2	1.12	1.33	0.51	3.37	81.1

CC-LA-$W_{0.5}$: Which was prepared by adding 0.5 mol of water to CC-LA.
CC-LA-W_1: Which was prepared by adding 1.0 mol of water to CC-LA.
CC-LA-W_2: Which was prepared by adding 2.0 mol of water to CC-LA.

3. Results and discussion

This work used three different acidic DESs to pretreat microalgae biomass. The solid–liquid ratio was 1:20, the pretreatment temperature was 80, 100, 120, and 140 °C, and the pretreatment time was 2, 4, 6, 8, and 10 h, respectively.The initial experiment studied the efficiency of DESs with different HBD for microalgae biomass pretreatment/fractionation. The selected DES (CC-LA) was added with an appropriate amount of water (molar mass of 0.5, 1.0, and 2.0) to study the effect of water on DES pretreatment efficiency. Finally, the components obtained from the optimal DES pretreatment of microalgae under optimal conditions were used for microbial fermentation, and the products were analyzed to evaluate the pretreatment effect.

3.1. The effect of DESs with different HBD on microalgae pretreatment

3.1.1. Synthesis and characterization of DESs

Three HBDs (FA, AA, and LA) were mixed with CC to obtain a colorless, uniform, and transparent liquid. FT-IR characterized the prepared DESs to determine the interaction between the components (Fig. 1). The pure and mixed components have obvious signals at 3000–3500 cm^{-1}, which is associated with the stretching vibration of O–H, indicating that a large number of hydrogen bonds are formed in DESs [28]. The C = O peaks of pure acids (FA, AA, and LA) were observed at 1702, 1706, and 1716 cm^{-1}, while the C = O peaks in the corresponding DESs showed a slight blue shift to 1712, 1716, and 1731

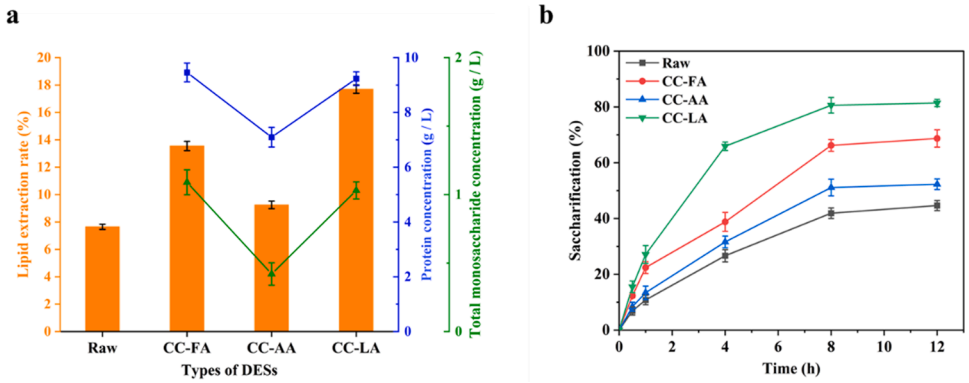


Fig. 2. (a) The fractionation efficiency of different DESs on microalgae biomass, and (b) the effect of different DESs on RMP saccharification rate.

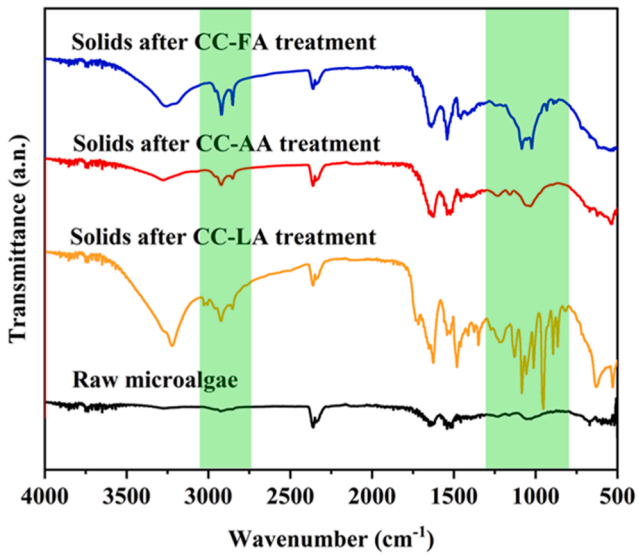


Fig. 3. FT-IR spectra of raw microalgae and RMP.

cm^{-1}, respectively [29]. This suggests that there is a strong interaction between different molecules of the mixed component.

The polarity of the solvent is an important parameter to evaluate its solvation ability, which will affect the reaction rate in the conversion process and its reaction mechanism [30]. Therefore, the solvatochromic probe (dye) of DESs was analyzed by a UV–visible spectrophotometer. K-T parameters usually describe the solvent polarity.. The values of π*, α, and β of the prepared DESs were quantified by N,N-dimethyl-4-nitrobenzene, Nile Red, and 4-nitroaniline dye staining to quantify the dipole/polarizability, hydrogen bond acidity, and hydrogen bond alkalinity, respectively, as shown in Table 1.

The π* values of the synthesized DESs are 1.18 (CC-FA), 0.99 (CC-AA), and 1.16 (CC-LA), respectively. A higher π* value indicates that DES has relatively higher polarizability [31], indicating that the solvents' polarity is in the order of CC-FA > CC-LA > CC-AA. The parameter α reflects the ability of the solvent to provide protons. The α values of DESs studied were between 1.19 and 1.37, with CC-LA showing the highest value (1.37) and CC-AA showing the lowest value (1.19). These three DESs all exhibit a high α value, that is, the superior ability to provide protons (acidity), considering that the HBDs of the synthesized DESs contain carboxylic acid groups. The parameter β reveals the ability of the solvent to accept protons (forming hydrogen bonds). The β values of the three DESs were basically stable at about 0.50, which were 0.47 (CC-FA), 0.51 (CC-AA), and 0.52 (CC-LA), respectively. However, the β values of these acidic DESs based on CC decreased in the HBD order of LA > AA > FA. The β value of DESs is positively correlated with the length of the HBD alkyl chain.

3.1.2. *The fractionation efficiency of DESs on microalgae pretreatment*

Subsequently, the prepared DESs were used for the pretreatment fractionation of microalgae, and the remaining solid (RMP) after pretreatment was utilized for enzymatic hydrolysis to evaluate the pretreatment effect. The fractionation of each component was shown in Fig. 2 after the pretreatment of microalgae with different DESs at 120 °C for 2 h. Unpretreated microalgae (Raw) only extracted 7.7 % lipid. The pretreatment of DESs effectively destroyed the microalgae cells and increased the solubilization of soluble substances (proteins, carbohydrates, and lipids) in microalgae cells. The extraction rate of lipids after pretreatment showed the following order: CC-LA (17.7 %) > CC-FA (13.6 %) > CC-AA (9.3 %). This trend was similar to the concentration of monosaccharides dissolved in liquid. Compared with CC-AA, CC-FA, and CC-LA had relatively higher lipid extraction rates and monosaccharide solubility because they have higher α values. DES with a high α value can release more protons, which can degrade the cell wall network more effectively [32]. The protein concentrations dissolved in the aqueous phase of DESs were 9.5 g/L (CC-FA), 7.1 g/L (CC-AA), and 9.2 g/L (CC-LA), respectively. The extraction rates of lipids (CC-LA > CC-FA > CC-AA) and proteins (CC-FA > CC-LA > CC-AA) by the three DESs were different, depending on their dissolution methods. The extraction efficiency of lipids is determined by the degree of cell damage. The higher the degree of damage, the more conducive to the extraction of ethyl acetate. The polarity and compatibility of the solvent determines the extraction rate of protein. The dissolution trend was consistent with the π* value of DESs, and the solvent with higher polarity contributes to the solubilization of proteins [12]. In general, the fractionation efficiency showed that CC-LA had a better pretreatment effect, followed by CC-FA and CC-AA, which was verified by the saccharification efficiency of RMP (Fig. 2b).

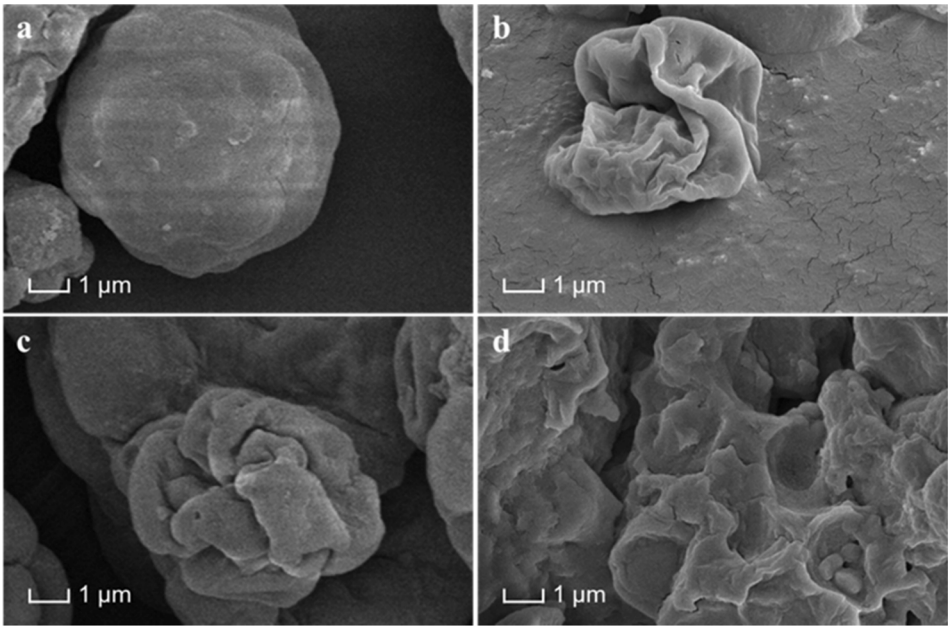


Fig. 4. SEM images of (a) raw microalgae; (b) RMP after CC-FA; (c) RMP after CC-AA; (d) RMP after CC-LA.

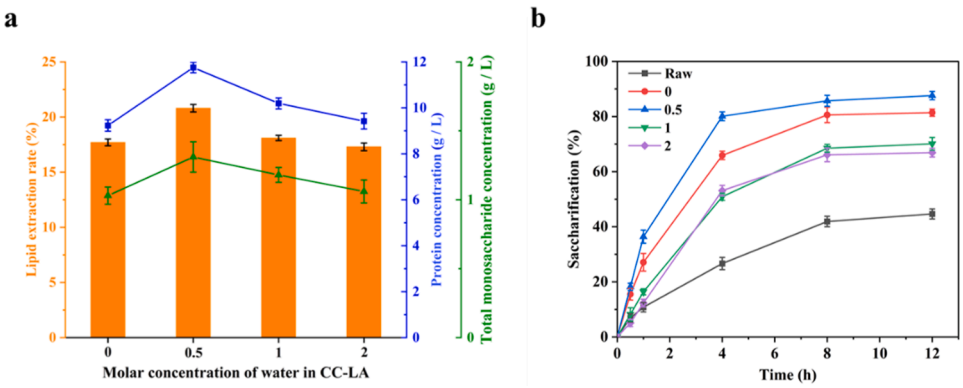


Fig. 5. (a) The fractionation efficiency of CC-LA with different water contents on microalgae biomass, and (b) the effect of CC-LA with different water content on RMP saccharification rate.

3.1.3. Characterization analysis of RMP

RMP was characterized by FT-IR (Fig. 3), showing that DESs effectively destroyed the cell structure of microalgae. The vibration bands around 800–1250 cm^{-1} and 2800–2950 cm^{-1} belong to the stretching of polysaccharides [33]. In all pretreated samples, the peak intensity of these regions was significantly improved, especially for CC-LA. This change can be explained by the efficient degradation of the microalgae cell wall network by DESs and more exposure to polysaccharides in the pretreated samples. The destructive effect of microalgae cells can be observed by SEM images of RMP (Fig. 4). Similarly, this exposure effect follows the order of CC-LA > CC-FA > CC-AA. The untreated microalgae cells were round and smooth without damage, while the cells treated with CC-FA and CC-AA showed shrinkage, and a small number of cracks were observed on CC-FA. The most obvious is CC-LA, where microalgae cells appeared with a large number of broken and observed massive cell debris.

As summarized in Table 1, the BET surface area of RMP after DESs pretreatment was higher (0.98–13.02 m^2/g) than that of untreated microalgae (0.66 m^2/g), which was increased by about 19.7 (CC-LA) times, 2.1 (CC-FA) times, and 1.5 (CC-AA) times, respectively. CC-LA shows a significant increase in surface area related to its higher α and β values. DESs with high α values easily provide hydrogen bonds to release H^+, while DESs with high β values will show the stronger ability to accept hydrogen bonds [32]. This DES with a high hydrogen supply and absorption capacity is more likely to reduce the hydrogen bond strength between microalgae cell walls, thereby achieving greater cell wall breakage [12]. This is consistent with the surface structure of RMP shown by SEM. The severe destruction of microalgae cells helps to increase the surface area, thereby increasing cellulose accessibility and promoting the saccharification of RMP.

The crystalline properties of RMP were analyzed by XRD (see supplementary information and Table 1). The diffraction pattern of RMP showed a signal similar to that of untreated microalgae, demonstrating that the form of carbohydrates before and after pretreatment did not change [34]. The pretreatment effect of DESs was evaluated by calculating the crystallinity index (CrI). The results showed that the CrI values of RMP components of microalgae biomass pretreated by various DESs increased from 72.3 % to 83.2 %, indicating that the pretreatment process improved the crystallinity of RMP. High crystallinity is closely related to the excellent enzymatic hydrolysis effect [35].

3.1.4. Enzymatic saccharification of RMP

Enzymatic saccharification efficiency is an important parameter to evaluate the pretreatment effect, which mainly includes the preparation of fermentable sugar [36]. As shown in Fig. 2b, after 12 h of enzymatic hydrolysis, the saccharification rate of CC-LA reached 81.4 %, followed by CC-FA of 68.7 %, while Raw and CC-AA were only 44.7 % and 52.3 %, respectively. Raw and CC-AA showed a lower saccharification rate, which may be attributed to the untreated cell wall or the degree of treatment (lysis) not ideal, resulting in a lower specific surface area, fewer enzyme sites, and a significantly reduced hydrolysis rate. CC-LA has a relatively high specific surface area (13.02 m^2/g), which exposes more active sites, significantly improving the probability of contact with the active site of the enzyme and thereby increasing the saccharification rate.

Table 2
Fatty acid composition of microalgae lipids and microbial lipids.

Fatty acids	Fatty acid composition (%, w/w)			
	Microalgae lipid	Microbial lipid (DES phase fermentation)	Microbial lipid (CK)	Microbial lipid (RMP)
Lauric acid (C12:0)	5.2	–	0.9	0.4
Myristic acid (C14:0)	30.1	–	1.8	–
Palmitic acid (C16:0)	49.5	20.2	20.4	20.7
Stearic acid (C18:0)	1.7	28.0	22.5	17.8
Oleic acid (C18:1)	11.2	44.8	46.0	57.0
Linoleic acid (C18:2)	2.2	7.0	8.4	4.1
Total (%)	100	100	100	100

3.2. Effect of water on the performance of DESs pretreated microalgae biomass

Most of the DESs are highly hydrophilic, increasing curiosity about aqueous DESs. In addition, adding water can effectively reduce the viscosity of DESs and make them exhibit better solvent properties [37]. Based on the above results, CC-LA was selected as the object, and an appropriate amount of water (0.5, 1.0, and 2.0 mol fractions) was added to it to study the effect of water on the pretreatment performance of DES.

With the increase of water content in DES, the K-T parameters (π*, α, and β) showed a trend of increasing first and then decreasing (Table 1). This is because adding a small amount of water makes it exist in the form of hydration in DES. In this case, water molecules bind to DES with stronger intermolecular interactions [38]. When the water content continues to increase, the DES system will repel water molecules to form a free state to maintain its internal structure. When the water content exceeds a specific value, the structure of DES is prone to be destroyed, and the molecular bonds, such as hydrogen bonds, are to decrease and

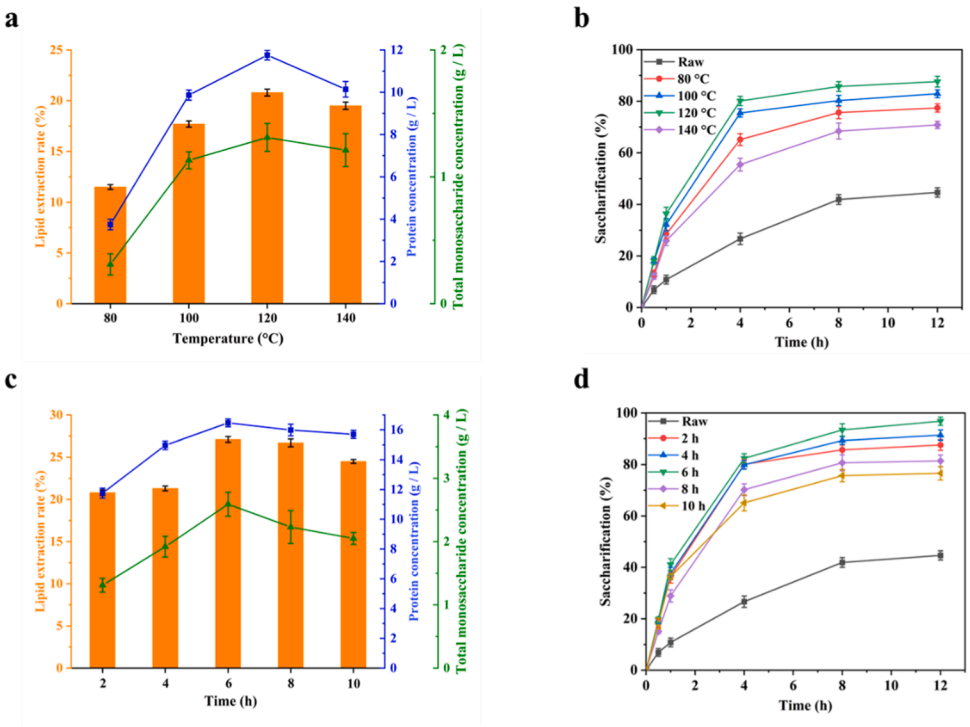


Fig. 6. The effect of CC-LA-$W_{0.5}$ on microalgae biomass fractionation at different (a) temperatures and (c) times, and the effect of different pretreatment (b) temperature and (d) time on RMP saccharification rate.

disappear [13]. Through K-T parameter analysis, it can be predicted that CC-LA-$W_{0.5}$ has the best pretreatment effect, which is also confirmed by pretreatment fractionation and enzymatic saccharification experiments (Fig. 5).

In general, it can be seen from Fig. 5a that the pretreatment effect increases first and then decreases with the continuous addition of water content. Among them, CC-LA-$W_{0.5}$ showed the best pretreatment effect (20.8 % lipid yield, 11.8 g/L protein, and 1.3 g/L monosaccharide). When the water content exceeded 0.5 mol (1.0 and 2.0 mol), the treatment efficiency began to decrease, and the efficiency was lower than that without water (CC-LA). This result corresponds to the K-T parameter results in Table 2. A small amount of water reduced the viscosity of DES, increased the number of hydrogen bonds (α value increased from 1.37 to 1.42), and improved the ability to accept hydrogen (β value from 0.52 to 0.54), thereby improving the ability of DES pretreatment fractionation. Excessive water destroyed the structure of DES and reduced the number of hydrogen bonds (α value was reduced to 1.33 when the water content was 2.0 mol) and the ability to accept hydrogen (β value was reduced to 0.51 when the water content was 2.0 mol), thereby reducing the pretreatment fractionation ability of DES [14].

Simultaneously, the corresponding RMP enzymatic saccharification efficiency was evaluated (Fig. 5b). The final saccharification efficiency of CC-LA-$W_{0.5}$ reached 87.6 %, higher than that of CC-LA (81.4 %). It could be attributed to the fact that adding an appropriate amount of water improves the pretreatment effect, which significantly increases the specific surface area of RMP (from 13.02 to 19.62 m^2/g, see Table 1). In addition, the decrease in the specific surface area leads to a decrease in saccharification efficiency, which confirms that the decrease in treatment efficiency of DES with high water content leads to low saccharification yield. Similarly, the trend of CrI values of RMP can also explain the trend of first increase and then decrease of saccharification.

3.3. The effects of CC-LA-$W_{0.5}$ pretreatment temperature and time on microalgae fractionation

The above studies have shown that CC-LA-$W_{0.5}$ has the best pretreatment fractionation effect, so the pretreatment performance of CC-LA-$W_{0.5}$ at different temperatures and times was further investigated (Fig. 6).

3.3.1. The effect of CC-LA-$W_{0.5}$ pretreatment temperature on microalgae fractionation and saccharification efficiency

As shown in Fig. 6a, the lipid extraction rate increased significantly from 11.5 % to 20.8 % after the pretreatment temperature increased from 80 °C to 120 °C. The amount of protein detected in the liquid phase of DES increased from 3.7 g/L to 11.8 g/L, and the dissolved monosaccharides also increased from 0.3 g/L to 1.3 g/L. Since the monosaccharides in the DES liquid phase are derived from the disintegrated microalgae cell wall and the proteins are derived from the cytoplasm, the increase in the concentration of monosaccharides and proteins indicates that increasing the temperature to 120 °C can promote the decomposition of microalgae biomass. However, when the temperature increased from 120 °C to 140 °C, the fractionation efficiency of each component showed a slight downward trend. Lipid, protein, and monosaccharide decreased to 19.5 %, 10.1 g/L, and 1.2 g/L, respectively. The involved reasons could be that lipids are easily degraded at high temperatures [39], and proteins and monosaccharides are more prone to undergo the Maillard reaction [40].

To measure the effect of temperature on the pretreatment of microalgae biomass, the enzymatic saccharification rate of RMP was investigated, as shown in Fig. 6b. At 80–120 °C, the saccharification efficiency was positively correlated with temperature, and the saccharification rate increased from 77.4 % to 87.6 %, which was about 1.7–2.0 times higher than that of raw. The significant increase in the saccharification rate was mainly attributed to the effective disintegration of the microalgae cell wall, which significantly improved the accessibility of the enzyme. However, when the temperature continued to rise to 140 °C, the saccharification rate decreased rapidly from 87.6 % to 70.8 %.

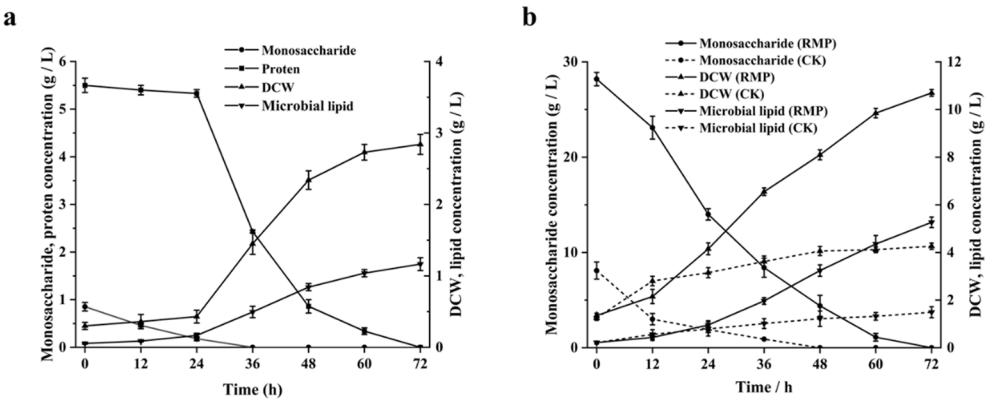


Fig. 7. *L. starkeyi* fermentation of (a) DES aqueous phase and (b) RMP enzymatic hydrolysate to produce microbial lipids.

Because a high temperature leads to side reactions, some toxic substances are adsorbed on the surface of RMP, thus inhibiting the saccharification rate of the enzyme [12,41].

3.3.2. The effect of CC-LA-$W_{0.5}$ pretreatment time on microalgae fractionation and saccharification efficiency

The effect of time on CC-LA-$W_{0.5}$ pretreatment is shown in Fig. 6c. In the initial reaction time of 4 h, the lipid extraction rate was stable at about 20 %, while the protein and monosaccharide showed a rapid increase from 11.8 g/L to 15.0 g/L and 1.3 g/L to 1.9 g/L, respectively. It shows that DES has a more obvious solubilization effect on protein and monosaccharide than lipids in a short time. The optimal effect was achieved at 6 h, when the extraction rate of lipids was 27.1 %, and the concentrations of protein and monosaccharides in the solution were 16.5 g/L and 2.6 g/L, respectively. However, the yields of the three products showed a downward trend when the pretreatment time was prolonged. After 10 h, the lipid extraction rate decreased to 24.5 %, and the protein and monosaccharide concentrations decreased to 15.7 g/L and 2.1 g/L, respectively. It is worth noting that long-term high-temperature treatment more likely leads to protein hydrolysis into amino acids, acylation reaction with lipids, and Maillard reaction with carbohydrates [42].

In the enzymatic saccharification experiment of RMP, as the pretreatment time increased to 6 h, the saccharification rate increased from 87.6 % (2 h) to 96.8 % (6 h), as shown in Fig. 6d, and decreased to 81.4 % and 76.6 % after 8 h and 10 h, respectively. The initial extension of time resulted in a complete lysis of microalgae cells, thereby increasing the saccharification rate. The longer time leads to the side reaction of the depolymerization product. Thus the toxic by-products inhibit the hydrolysis efficiency of the enzyme and reduce the saccharification rate.

3.4. Lipid production from microalgae pretreated by the CC-LA-$W_{0.5}$

The ultimate goal of pretreatment is to use biomass components more efficiently. Therefore, the fermentation feasibility of DES aqueous phase and RMP enzymatic hydrolysate after CC-LA-$W_{0.5}$ pretreatment (120 °C for 6 h) was explored, as shown in Fig. 7.

3.4.1. Fermentation of DES aqueous phase by L. Starkeyi

The aqueous phase of DES was diluted three times and used as a nutrient source to produce microbial lipids, mainly using proteins and small amounts of monosaccharides dissolved in DES. As shown in Fig. 7a, the lag phase of *L. starkeyi* was observed in the first 24 h, showing slow cell growth (a slight increase in DCW). Because of the DES system, the fermentation broth contains more or fewer fermentation inhibitors to inhibit cell growth, thus prolonging the lag phase of yeast [43]. In addition, the protein was almost not consumed during this period, mainly using monosaccharides, which was in line with the order of microbial utilization of carbon sources (Reducing sugars present in the system were preferentially utilized by microorganisms) [44]. Subsequently, *L. starkeyi* entered the exponential growth phase, the cells grew rapidly and accumulated lipids and proteins, and monosaccharides were rapidly consumed. It is worth noting that monosaccharides were not detected at 36 h. Finally, 2.8 g/L DCW and 1.2 g/L microbial lipids were obtained after 72 h fermentation. This result indicates that the protein and a small amount of monosaccharides of microalgae are fractionated in the liquid phase after DES pretreatment, which can be efficiently utilized by biological fermentation.

3.4.2. Fermentation of RMP enzymatic hydrolysate by L. Starkeyi

The enzymatic hydrolysate of RMP (solid loading of 5 %) was employed for *L. starkeyi* fermentation to produce microbial lipids, and the enzymatic hydrolysate of untreated microalgae was used as a control (CK). In Fig. 7b, the initial monosaccharide concentrations were 28.2 g/L (RMP) and 8.1 g/L (CK), respectively. This is preferred for the use of solid enzymatic hydrolysates as a carbon source, as no toxic inhibitors exist in the fermentation broth to ensure the normal growth of *L. starkeyi*. Therefore, it can be observed that there is almost no lag phase in the whole process, and the cells grow rapidly. In the first 12 h, the RMP (2.2 g/L DCW) group showed slower cell growth than the CK (2.8 g/L DCW) group. Because of the high concentration of monosaccharides in the fermentation broth of the RMP group, the cells need to adapt to the high-concentration environment for a while, thus stagnating growth resulting in a lower growth rate than the CK group. After 12 h, the cells in the RMP (4.2 g/L DCW) group proliferated, while the growth rate of the CK (3.2 g/L DCW) group slowed down, and there was almost no growth after 48 h. Because there were few monosaccharides left in the fermentation broth of the CK group after the first 12 h of fermentation, only 3.0 g/L monosaccharides were left, and the monosaccharides were exhausted after 48 h. Finally, after 72 h of fermentation, the monosaccharides in the RMP group were completely utilized by yeast. After 72 h, 10.7 g/L (RMP) and 4.3 g/L (CK) DCW, 5.3 g/L (RMP), and 1.5 g/L (CK) microbial lipids were obtained, respectively. These results demonstrate that the DES pretreatment can effectively improve the bioavailability of carbohydrates in microalgae.

3.5. Fames composition analysis

The composition of fatty acids is the main factor affecting the quality of biodiesel, so the composition of microalgae lipids and microbial lipids obtained by fermentation was analyzed (Table 2). All the obtained fatty acids mainly include lauric acid (C12:0), myristic acid (C14:0), palmitic acid (C16:0), stearic acid (C18:0), oleic acid (C18:1), and linoleic acid (C18:2). The fatty acid composition of microalgae lipids fractionated by pretreatment showed that myristic acid (C14:0), palmitic acid (C16:0), and oleic acid (C18:1) accounted for 30.1 %, 49.5 %, and 11.2 % of the

Table 3
Lipid production by yeast from different microalgae biomass and fatty acid composition.

strain	substrate	Lipids (g/L)	Main fatty acid composition	reference
C. oleaginosus	*S. obtusiusculus* hydrolysates	3.6	–	[47]
C. curvatus	*A. platensis* hydrolysates	3.8	Palmitic acid (C16:0)Oleic acid (C18:1)Stearic acid (C18:0)	[49]
C. curvatus	*A. platensis* hydrolysates with phosphate removal	9.6	Palmitic acid (C16:0)Oleic acid (C18:1)Stearic acid (C18:0)	[49]
T. oleaginosus	*P. oceanica* hydrolysates	6.8	Palmitic acid (C16:0) Stearic acid (C18:0) Oleic acid (C18:1)	[48]
Nannochloropsis sp.	*L. starkeyi*	6.5	Palmitic acid (C16:0) Stearic acid (C18:0) Oleic acid (C18:1)	this work

total fatty acids, respectively. The fatty acids in microbial lipids obtained by fermentation were mainly palmitic acid (C16:0), stearic acid (C18:0), and oleic acid (C18:1). The higher the content of oleic acid and palmitic acid, the better the quality of biodiesel is [44]. It can be seen from Table 3 that the quality of microbial lipids is better than that of microalgae lipids. It is a wise choice to convert other biomass of microalgae except for lipids into microbial lipids. In addition, the higher the C/N ratio in the fermentation medium (RMP > CK > DES), the higher the content of palmitic acid and oleic acid in microbial lipids, and the less the content of stearic acid were observed. It is indicated that a high C/N ratio is beneficial to obtain higher-quality microbial lipids.

3.6. Mass balance

100 g of initial microalgae was used to calculate the mass balance, and the difference in conversion efficiency between untreated and CC-LA-$W_{0.5}$ pretreated microalgae was compared, as shown in Fig. 8. That every 100 g of dry microalgae contains 15.3 g of carbohydrates, 39.2 g of proteins and 30.3 g of lipids. The direct extraction of microalgae lipids from untreated microalgae was only 15.3 g (50.5 %). For the pretreatment group, the cells were effectively destroyed after the pretreatment of microalgae so that the two could be fully contacted when using ethyl acetate to extract microalgae lipids, which significantly improved the extraction rate. Finally, 27.1 g (89.4 %) of microalgae lipids were obtained. Due to their relatively complete cell wall structure, the remaining solids in the untreated group resulted in only 6.9 g of carbohydrates produced during the enzymatic hydrolysis process. Finally, 1.2 g microbial lipid was produced by fermentation. Unlike the untreated group, the pretreated DES contained a large amount of protein (33.0 g) and a small amount of carbohydrate (5.2 g), which could produce 7.0 g of microbial lipids by fermentation. In the solid part, the cell wall was fully broken, which was conducive to the enzyme hydrolysis, and 9.8 g of carbohydrate was obtained. Moreover, 9.8 g carbohydrates were fermented into 1.9 g microbial lipids. Therefore, 8.9 g microbial lipids were obtained by fermentation of the final biomass in the pretreatment group. In summary, the pretreatment step dramatically improves the conversion efficiency of microalgae biomass, making it a potential source of biodiesel.

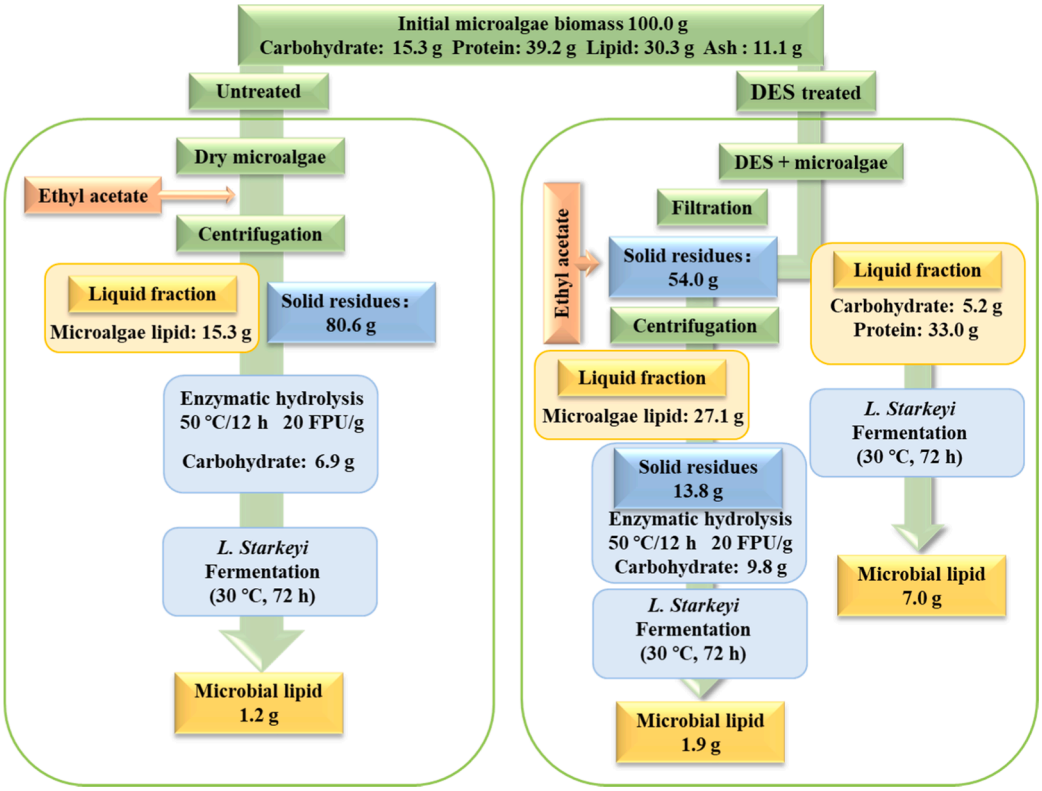


Fig. 8. Untreated and pretreated microalgae biomass material balance.

3.7. Prospect of lipid production from microalgae biomass

Microalgae have been used for many years as a source of value-added products with many commercial applications. The significant characteristics of microalgae biomass as a raw material for microbial culture include high carbohydrate content and lack of stubborn lignin [45,46]. Based on this, many studies have used microalgae biomass as fermentation raw materials to obtain more valuable microbial lipids (Table 3). For example, 3.6 g/L microbial lipid was obtained using the hydrolysate of *Scenedesmus obtusiusculus* as the fermentation carbon source [47]. *Trichosporon oleaginosus* obtained 6.8 g/L microbial lipids from the hydrolysate of *P. oceanica* [48]. The hydrolysate of *Arthrospira platensis* was fermented to obtain 9.6 g/L microbial lipids [49]. However, these biological pretreatment processes use expensive enzymatic combined treatment or chemical acid heat treatment. Although some have obtained considerable lipid yields, redundant detoxification steps, and high costs make it challenging to use microalgae biomass as a source of biofuels. Meanwhile, only carbohydrates are used, and most proteins are wasted.

In this study, the hydrolysate can be directly used for fermentation to obtain 6.5 g/L microbial lipids without additional detoxification steps. All components of microalgae biomass were fully utilized, and proteins and carbohydrates were converted into higher-quality microbial lipids (mainly palmitic acid, stearic acid, and oleic acid). This provides excellent help for the efficient utilization of microalgae biomass.

4. Conclusions

A novel water-based DES pretreatment strategy was proposed to efficiently fractionate microalgae biomass and achieve full-component utilization of microalgae biomass. The CC-LA was illustrated to exhibit better pretreatment efficiency for more thoroughly breaking the microalgae cells, and a high saccharification rate of up to 83.2 % was obtained. The specific surface area and crystallinity of CC-LA can be significantly improved by adding the appropriate amount of water. Under the optimal pretreatment conditions, the extraction rate of microalgae lipid reached 89.4 %, and the saccharification rate of RMP achieved 96.8 %. The protein and carbohydrate obtained by fractionation could be fermented to 8.9 g microbial lipid/100 g microalgae. The CC-LA-$W_{0.5}$ prepared in this study is a new type of pretreatment solvent and can efficiently utilize each component of microalgae.

CRediT authorship contribution statement

Dayong Yu: Writing – original draft. **Hangyu Luo:** Investigation. **Jinyu Tan:** Investigation. **Junfa Yuan:** Investigation. **Jiasheng Chen:** Investigation. **Xiaofang Liu:** . **Hu Li:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2024.130892.

References

[1] Wang Y, Liu X, Liu Y, Wang D, Xu Q, Li X, et al. Enhancement of short-chain fatty acids production from microalgae by potassium ferrate addition: Feasibility, mechanisms and implications. Bioresour Technol 2020;318:124266.
[2] Chisti Y. Biodiesel from microalgae. Biotechnol Adv 2007;25(3):294–306.
[3] Huang J, Wang J, Huang Z, Liu T, Li H. Photothermal technique-enabled ambient production of microalgae biodiesel: Mechanism and life cycle assessment. Bioresour Technol 2023;369:128390.
[4] Barati B, Zafar FF, Rupani PF, Wang S. Bacterial pretreatment of microalgae and the potential of novel nature hydrolytic sources. Environ Technol Innov 2021;21:101362.
[5] Bernaerts TMM, Gheysen L, Foubert I, Hendrickx ME, Van Loey AM. Evaluating microalgal cell disruption upon ultra high pressure homogenization. Algal Res 2019;42:101616.
[6] Kumar V, Nanda M, Joshi HC, Singh A, Sharma S, Verma M. Production of biodiesel and bioethanol using algal biomass harvested from fresh water river. Renew Energ 2018;116:606–12.
[7] de Carvalho JC, Magalhaes Jr AI, de Melo Pereira GV, Medeiros ABP, Sydney EB, Rodrigues C, et al. Microalgal biomass pretreatment for integrated processing into biofuels, food, and feed. Bioresour Technol 2020;300:122719.
[8] Jose D, Kitiborwornkul N, Sriariyanun M, Keerthi K. A Review on Chemical Pretreatment Methods of Lignocellulosic Biomass: Recent Advances and Progress. Applied Science and Engineering Progress 2022;15(4):6210.
[9] Jose D, Tawai A, Divakaran D, Bhattacharyya D, Venkatachalam P, Tantayotai P, et al. Integration of deep eutectic solvent in biorefining process of lignocellulosic biomass valorization. Bioresource Technology Reports 2023;(21):101365.
[10] Shu F, Guo Y, Huang L, Zhou M, Zhang G, Yu H, et al. Production of lignin-containing nanocellulose from poplar using ternary deep eutectic solvents pretreatment. Ind Crop Prod 2022;177:114404.
[11] Show P-L, Sriariyanun M. Prospect of Liquid Biphasic System in Microalgae Research. Applied Science and Engineering Progress 2020;14(3):295–6.
[12] Huang R, He Y, Yao X, Yu Y, Song W, Yang W, et al. Disintegration of wet microalgae biomass with deep-eutectic-solvent-assisted hydrothermal treatment for sustainable lipid extraction. Green Chem 2022;24(4):1615–26.
[13] Hammond OS, Bowron DT, Edler KJ. The Effect of Water upon Deep Eutectic Solvent Nanostructure: An Unusual Transition from Ionic Mixture to Aqueous Solution. Angew Chem Int Edit 2017;56(33):9782–5.
[14] Chourasia VR, Bisht M, Pant KK, Henry RJ. Unveiling the potential of water as a co-solvent in microwave-assisted delignification of sugarcane bagasse using ternary deep eutectic solvents. Bioresour Technol 2022;351:127005.
[15] Martani F, Maestroni L, Torchio M, Ami D, Natalello A, Lotti M, et al. Conversion of sugar beet residues into lipids by *Lipomyces starkeyi* for biodiesel production. Microb Cell Fact 2020;19(1):204.
[16] Sutanto S, Zullaikah S, Tran-Nguyen PL, Ismadji S, Ju Y-H. *Lipomyces starkeyi*: its current status as a potential oil producer. Fuel Process Technol 2018;177:39–55.
[17] Kong L, Fan B, He Y-C. Efficient whole-cell biosynthesis of (*S*)-2-chloro-1-(3,4-difluorophenyl)-ethanol from 2-chloro-1-(3,4-difluorophenyl) ethanone in a sustainable reaction system. Mol Catal 2023;550:113570.
[18] Liu Y, Li L, Ma C, He YC. Chemobiocatalytic transfromation of biomass into furfurylamine with mixed amine donor in an eco-friendly medium. Bioresour Technol 2023;387:129638.
[19] Van Wychen S, Laurens LML. Determination of total carbohydrates in algal biomass: laboratory analytical procedure (LAP). NREL/TP-5100-60957 National Renewable Energy Lab (NREL) 2016;Colden:CO (United States).
[20] Lowry O, Rosebrough N, Farr AL, Randall R. Protein Measurement with the Folin Phenol Reagent. J Biol Chem 1951;193(1):265–75.
[21] Bligh EG, Dyer WJ. A rapid method of total lipid extraction and purification. Can J of Bio Physiol 1959;37:911–7.
[22] Di Fidio N, Ragaglini G, Dragoni F, Antonetti C, Raspolli Galletti AM. Integrated cascade biorefinery processes for the production of single cell oil by *Lipomyces starkeyi* from *Arundo donax* L. hydrolysates. Bioresour Technol 2021;325:124635.
[23] Tasselli G, Filippucci S, Borsella E, D'Antonio S, Gelosia M, Cavalaglio G, et al. Yeast lipids from cardoon stalks, stranded driftwood and olive tree pruning residues as possible extra sources of oils for producing biofuels and biochemicals. Biotechnol Biofuels 2018;11:147.
[24] Chatterjee S, Mohan SV. Microbial lipid production by *Cryptococcus curvatus* from vegetable waste hydrolysate. Bioresour Technol 2018;254:284–9.
[25] Liang X, Zhu Y, Qi B, Li S, Luo J, Wan Y. Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment. Bioresour Technol 2021;336:125312.
[26] Li X, Tang W, He Y-C. Integrated understanding of acidic deep eutectic solvent choline chloride: Oxalic acid pretreatment to enhance the enzymatic hydrolysis of rape straw. Ind Crop Prod 2023;206:117691.

[27] Mary Leema JT, Persia Jothy T, Dharani G. Rapid green microwave assisted extraction of lutein from *Chlorella sorokiniana* (NIOT-2) - Process optimization. Food Chem 2022;372:131151.
[28] Gajardo-Parra NF, Lubben MJ, Winnert JM, Leiva Á, Brennecke JF, Canales RI. Physicochemical properties of choline chloride-based deep eutectic solvents and excess properties of their pseudo-binary mixtures with 1-butanol. J Chem Thermodyn 2019;133:272–84.
[29] Zhu J, Shao H, Feng L, Lu Y, Meng H, Li C. Absorptive separation of HCl gas by choline chloride-based deep eutectic solvents. J Mol Liq 2021;341:116928.
[30] Katritzky AR, Fara DC, Yang H, Tämm K, Tamm T, Karelson M. Quantitative Measures of Solvent Polarity. Chem Rev 2004;104(1):175–98.
[31] Kamlet MJ, Abboud JL, Taft RW. The solvatochromic comparison method. 6. The. pi.* scale of solvent polarities. J Am Chem Soc 1977;99(18):6027–38.
[32] Xia Q, Liu Y, Meng J, Cheng W, Chen W, Liu S, et al. Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass. Green Chem 2018;20(12):2711–21.
[33] Grace CEE, Lakshmi PK, Meenakshi S, Vaidyanathan S, Srisudha S, Mary MB. Biomolecular transitions and lipid accumulation in green microalgae monitored by FTIR and Raman analysis. Spectrochim Acta A 2020;224:117382.
[34] Zhong L, Wang C, Yang G, Chen J, Xu F, Geun Yoo C, et al. Rapid and efficient microwave-assisted guanidine hydrochloride deep eutectic solvent pretreatment for biological conversion of castor stalk. Bioresour Technol 2022;343:126022.
[35] Wang Z-K, Li H, Lin X-C, Tang L, Chen J-J, Mo J-W, et al. Novel recyclable deep eutectic solvent boost biomass pretreatment for enzymatic hydrolysis. Bioresour Technol 2020;307:123237.
[36] Ma CY, Xu LH, Zhang C, Guo KN, Yuan TQ, Wen JL. A synergistic hydrothermal-deep eutectic solvent (DES) pretreatment for rapid fractionation and targeted valorization of hemicelluloses and cellulose from poplar wood. Bioresour Technol 2021;341:125828.
[37] Bish M, Martins M, Dias ACRV, Ventura SPM, Coutinho JAP. Uncovering the potential of aqueous solutions of deep eutectic solvents on the extraction and purification of collagen type I from Atlantic codfish (*Gadus morhua*). Green Chem 2021;23(22):8940–8.
[38] Francisco M, van den Bruinhorst A, Kroon MC. Low-transition-temperature mixtures (LTTMs): a new generation of designer solvents. Angew Chem Int Edit 2013;52(11):3074–85.
[39] Obeid R, Lewis DM, Smith N, Hall T, van Eyk P. Reaction kinetics and characterisation of species in renewable crude from hydrothermal liquefaction of monomers to represent organic fractions of biomass feedstocks. Chem Eng J 2020;389:124397.
[40] Wu H, Li J, Yang H, Liao Q, Fu Q, Liu Z. Hydrothermal treatment of *Chlorella* sp.: Influence on biochemical methane potential, microbial function and biochemical metabolism. Bioresour Technol 2019;289.
[41] Shen X-J, Wen J-L, Mei Q-Q, Chen X, Sun D, Yuan T-Q, et al. Facile fractionation of lignocelluloses by biomass-derived deep eutectic solvent (DES) pretreatment for cellulose enzymatic hydrolysis and lignin valorization. Green Chem 2019;21(2):275–83.
[42] He S, Zhao M, Wang J, Cheng Z, Yan B, Chen G. Hydrothermal liquefaction of low-lipid algae *Nannochloropsis* sp. and *Sargassum* sp.: Effect of feedstock composition and temperature. Sci Total Environ 2020;712.
[43] Soares LB, Bonan CIDG, Biazi LE, Dionísio SR, Bonatelli ML, Andrade ALD, et al. Investigation of hemicellulosic hydrolysate inhibitor resistance and fermentation strategies to overcome inhibition in non-saccharomyces species. Biomass Bioenerg 2020;137:105549.
[44] El-Dalatony MM, Salama E-S, Kurade MB, Kim K-Y, Govindwar SP, Kim JR, et al. Whole conversion of microalgal biomass into biofuels through successive high-throughput fermentation. Chem Eng J 2019;360:797–805.
[45] Baudelet P-H, Ricochon G, Linder M, Muniglia L. A new insight into cell walls of Chlorophyta. Algal Res 2017;25:333–71.
[46] Rashidi B, Trindade LM. Detailed biochemical and morphologic characteristics of the green microalga *Neochloris oleoabundans* cell wall. Algal Res 2018;35:152–9.
[47] Younes S, Bracharz F, Awad D, Qoura F, Mehlmer N, Brueck T. Microbial lipid production by oleaginous yeasts grown on *Scenedesmus obtusiusculus* microalgae biomass hydrolysate. Bioprocess Biosyst Eng 2020;43(9):1629–38.
[48] Masri MA, Younes S, Haack M, Qoura F, Mehlmer N, Brück T. A Seagrass-Based Biorefinery for Generation of Single-Cell Oils for Biofuel and Oleochemical Production. Energy Technol 2017;6(6):1026–38.
[49] Kamal R, Huang Q, Li Q, Chu Y, Yu X, Limtong S, et al. Conversion of *Arthrospira platensis* Biomass into Microbial Lipids by the Oleaginous Yeast *Cryptococcus curvatus*. ACS Sustain Chem Eng 2021;9(33):11011–21.